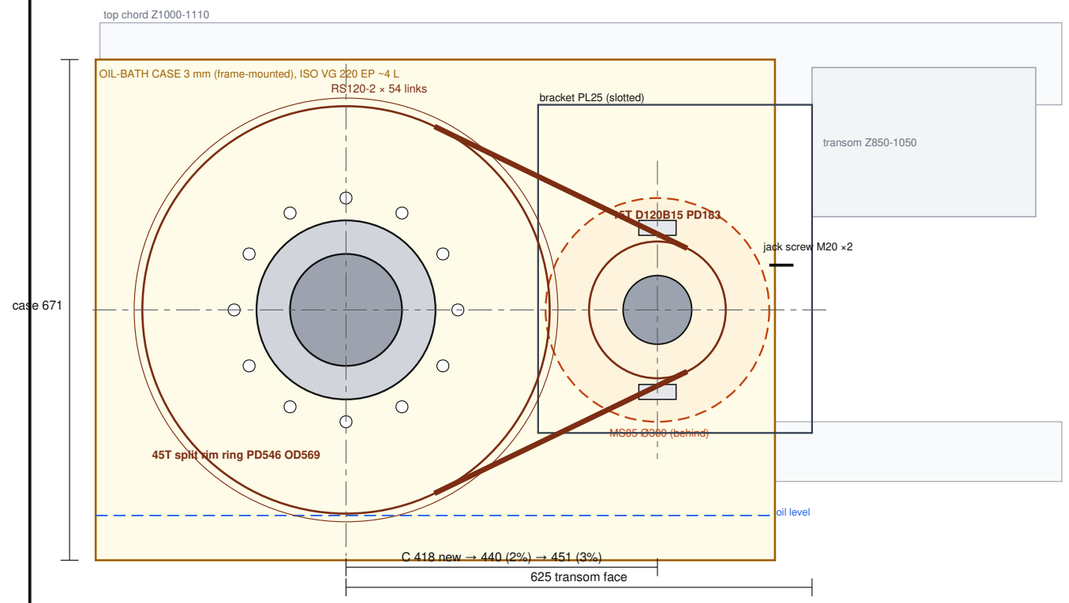
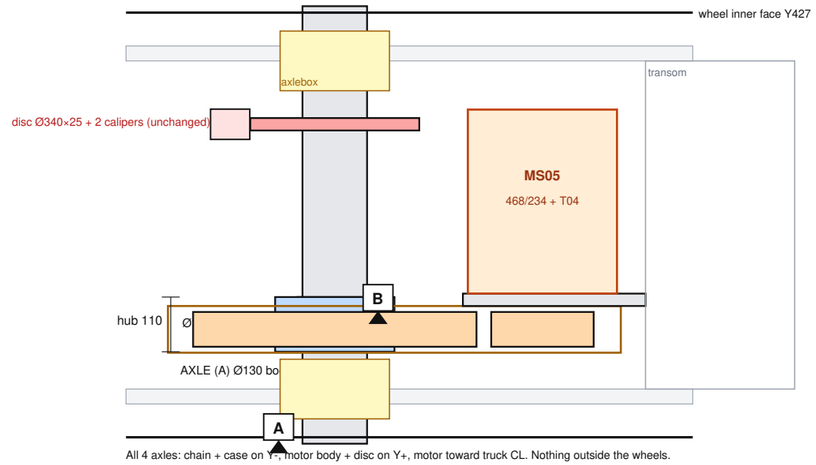
VIEW A - SIDE (from Y-, chain case cover removed; truck centre to the right)



Motor at 3% stretch clears the transom by 23 mm; case bottom Z389 ARL; chain line horizontal (±25 mm travel → +0.75 mm centres): set sag 17-20 mm, no idler.

SECTION B-B 1:12 (plan through axle CL; chain/motor flange Y-, motor body + disc Y+)



All 4 axles: chain + case on Y+, motor body + disc on Y+, motor toward truck CL. Nothing outside the wheels.
Seal plate: lip seal on hub land, PTFE-faced plate slides ±25 mm vertical / ±3 lateral in a case guide (OPEN OI-L4-5).

BILL OF MATERIALS - ONE DRIVE UNIT (×4 per loco)

Item	Description / material	Qty	Source
1	Poclain MS05 2-disp. 468/234 cc HighFlow, T04 brake, shaft code 5 DIN 5480 W55x3x17, speed sensor 56 ppr	1	AWAIT Poclain
2	Sprocket 15T 120-2 OD 202, MADE-TO-ORDER 4140 Q&T hub (L4.2), DIN 5480 N55x3x17, ≥ 58 engaged	1	AWAIT maker
3	Sprocket 45T 120-2 rim-only SPLIT ring, 2 halves (~50 kg, ASSUMED), 12 × M16 8.8 fitted, 219 N·m, to Ø350 hub flange; replace in place	1	AWAIT supplier
4	Hub 4140 Q&T Ø240 × 110 + Ø350 × 20 flange (31.4 kg), bore Ø150, SHRINK ONLY 0.15-0.2 at 200-230 °C; removal oil groove	1	machine shop
5	Chain ANSI 120-2, mill cert. MBL ≥ 326 kN per lot (L4.2; std RS120-2 296 kN NOT accepted), 54 links + conn. link	1	AWAIT chain maker
6	Oil-bath case 3 mm + floating seal plate + lip seal, sight glass, breather, mag. drain	1	fab / AWAIT seal
7	Motor bracket PL25 A572-50, slotted (4 × M20 10.9 in 50 slots), 2 jack screws M20, gussets	1	fab
8	Axle 34CrNiMo6 Q&T, Ø150 raised seat (no keyway, no hardening credit), UT + MT	1	machine shop
9	Disc Ø340×25 + 2 spring-applied air-released calipers (unchanged)	1+2	AWAIT brake sup.

GD&T (ASME Y14.5-2018)

CYL	0.008	-	Axle sprocket seat Ø150 (DATUM B) - hub bore for 0.15-0.2 mm diametral interference, Ra 0.8
↗	0.020	A	Seat radial runout to bearing journals (DATUM A)
↗	0.03	A	Seat shoulders r13 + undercut at hub edge; NO keyway; oil-injection groove + M8 port
↗	0.25	B	45T plate pitch circle runout to hub (fitted bolts PCD 300)
▱	0.40	B	15T and 45T tooth faces coplanar (chain alignment)
⌷	0.30	A	Motor slide plate square to axle in plan (slots parallel to chain line)
↗	0.05	A	Disc seat - runout

NOTES

1. 2% start: need 32.61 kN; 37.67 kN @350 bar (η 0.90). NO divider: PLUS+1 traction control; equal torque 1.14 dry/sanded, 1.146 with truck pitch (traction pin h 0.65; F2 37.11 kN, ballast 2 × 75.4 kg). SAND mandatory wet.
2. L4.2: chain pull 35.2 kN at the 430 bar strength basis (HPRV 380) → certified MBL ≥ 326 kN (factor 9.24); 296 kN std chain NOT accepted.
3. Seat Ø150, NO hardening credit, fretting Kf 2.0, heaviest axle 49.12 kN: Goodman n 1.57 ≥ 1.5. Fit slip 29.9 kN·m = 3.11× HPRV.
4. GOVERNED 6 mph: stroke stop 42 cc sealed + HARD tamper-sealed 2,600 rpm → 5.96 mph in 2nd; indep. 7 mph trip; hardware max 7.64 mph.
5. Single-axle spin 421 rpm at high idle (2nd) vs HighFlow 430 → HighFlow REQUIRED + traction control.
6. Motor radial 35.2 kN vs ~60 kN (code 5 chart) - sprocket close to flange; AWAIT Poclain AE. 2A40 keyed REJECTED.
7. Upkeep: oil weekly, change 500 h/season; sag monthly (17-20 mm at ride height); stretch 250 h; REPLACE CHAIN AT 2% (hard). Re-shim at centre plate only; primary ≤ 3 mm + slide reset. Unsprung drive 127 kg/axle.
8. T04 = parking only; L4.2 release from the H1 CHARGE circuit (M3) via 20 bar reducer (loop shuttle deleted); Ø1.2 orifice per port, hard/4SH lines; 2-ch switch vents BP first.
9. Chain: cert. MBL ≥ 326 kN per lot; bump stops ±22; reversal only at zero speed + 1 s ramp. Descents in FLOAT (loop bypass, ≥ 3 s ramp).
10. Wheels: condemn drivers at Ø1430; flange height WITHDRAW at 28 mm, re-profile ≤ 27 (train-wide, L4.2); enter diameters in PLUS+1.

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Final drive unit (L-103 sh 2) - Poclain MS05 2-speed + ANSI 120-2 15T:45T, 4140 hub (GD&T)

Sheet L-103.2 · REV L4.2 · 2026-10-28 · 1:8 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED